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Cardiac Rehabilitation for Patients With Heart Failure

JACC Expert Panel

Perspective From the ACC Heart Failure and Transplant Section and Leadership Council, Reviewed by the Prevention Section Cardiac Rehabilitation Work Group

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ABSTRACT

Cardiac rehabilitation is defined as a multidisciplinary program that includes exercise training, cardiac risk factor modification, psychosocial assessment, and outcomes assessment. Exercise training and other components of cardiac rehabilitation (CR) are safe and beneficial and result in significant improvements in quality of life, functional capacity, exercise performance, and heart failure (HF)-related hospitalizations in patients with HF. Despite outcome benefits, cost-effectiveness, and strong practice guideline recommendations, CR remains underused. Clinicians, health care leaders, and payers should prioritize incorporating CR as part of the standard of care for patients with HF. (J Am Coll Cardiol 2021;77:1454–69) Published by Elsevier on behalf of the American College of Cardiology Foundation.

Historically, patients with heart failure (HF) were assumed to be at risk to exercise and were commonly discouraged from participating in physical activity. Contrary to these concerns, multiple studies have demonstrated the safety and benefits of exercise and physical activity in patients with HF (1-4) and deleterious effects of prolonged bed rest and immobilization (5,6). The

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HIGHLIGHTS

- CR is underused among patients with HF.
- ET can improve functional capacity and clinical outcomes in patients with HF.
- Clinicians and payers should prioritize CR in the routine care of patients with HF.

HF-ACTION (Heart Failure: A Controlled Trial Investigating Outcomes of Exercise Training) trial, the largest trial of exercise training (ET) in patients with HF with reduced ejection fraction (HFrEF) (1), demonstrated that regular aerobic ET was very well tolerated, was safe, and improved quality of life (QOL). There were modest reductions in all-cause mortality and hospitalization rates, which did not reach significance by primary analysis but, after prespecified adjustment, appeared to be associated with reductions in cardiovascular mortality or HF hospitalizations (1). In a post hoc analysis of patients adherent to ET, the endpoints became strongly significant (7,8). The HF-ACTION trial further confirmed the benefits and safety of ET across several subgroups regardless of age, etiology of HF, severity of HF, race, and sex (1,9). Subsequent meta-analyses have further confirmed the safety (4,10) and efficacy of ET in HF, with significant improvements in QOL in both patients with HFrEF and those with HF with preserved ejection fraction (HFpEF) (11-16) and reductions in HF hospitalizations in patients with HFrEF (11,12,17).

Despite such encouraging results and cost-effectiveness, cardiac rehabilitation (CR) remains underused (18-20). Underuse stems from multiple factors, among which physician referral, payer coverage, circumscribed coverage criteria to patients on treatment for at least 6 weeks, patient adherence, and lack of awareness and education about its benefits are key factors (18-20). In this paper, we review the existing evidence and guidelines and provide guidance for the implementation of CR in patients with HF.

ETIOLOGY AND REVERSIBILITY OF EXERCISE INTOLERANCE IN PATIENTS WITH HF

Exercise intolerance, chronic fatigue, and inability to perform activities are cardinal manifestations of HF and are associated with poor QOL (21) and adverse outcomes (22). Independent of left ventricular function, patients with better exercise performance have lower mortality and hospitalization rates (23).

The reasons for exercise intolerance in patients with HF are multifactorial and include central cardiac and peripheral mechanisms (24-28). There is evidence of inadequate cardiac output and high filling pressures with an insufficient increase in perfusion to exercising muscle, leading to early anaerobic metabolism and muscle fatigue, in patients with HF (29-31). Skeletal muscle dysfunction manifested by impaired peripheral oxygen extraction and alterations in fiber composition, contractile efficiency, and metabolism also play roles (25,27,32-35). Other factors include endothelial dysfunction, obesity, increased sympathetic activation, vasoconstriction, and increased levels of inflammatory cytokines (36,37). There are likely differences in the pathophysiology of exercise intolerance in patients with HFrEF compared with those with HFpEF. Chronotropic incompetence is likely to play a key role in limiting exercise in patients with HFpEF.

Trials in patients with HF examining ET have shown reversal or attenuation of neurohormonal and inflammatory activation and ventricular remodeling (38-41). ET has also been associated with improvement in vasomotor and endothelial function, skeletal muscle morphological characteristics and function, ventricular filling pressures, exercise performance, and QOL in patients with HF (2,3,15,38-50) (**Central Illustration**).

DEFINITION AND COMPONENTS OF A CR PROGRAM

CR is defined as “a physician-supervised program that furnishes physician prescribed exercise, cardiac risk factor modification, psychosocial assessment, and outcomes assessment” by commonly accepted terminology and federal statute (51).

A CR program in HF is a comprehensive intervention with several components. In addition to ET, it includes patient assessment, education about medication adherence, risk factor modification including dietary recommendations, lifestyle modification, smoking cessation counseling, stress management, and evaluation and management of barriers to adherence. CR has evolved over the past 3 decades from a monitored exercise program to a comprehensive and multidisciplinary program (52-54).

Although the types and qualifications of CR team members may vary from program to program, a multidisciplinary CR team typically includes a physician medical director (55), nurses, advanced practitioners, exercise specialists or physiologists, and dietitians who work in collaboration and communication with referring providers, patients, and families

ABBREVIATIONS AND ACRONYMS

CR = cardiac rehabilitation

ET = exercise training

FITT = frequency, intensity, time, and type

HF = heart failure

HFrEF = heart failure with reduced ejection fraction

HFpEF = heart failure with preserved ejection fraction

HIIT = high-intensity interval training,

LVEF = left ventricular ejection fraction

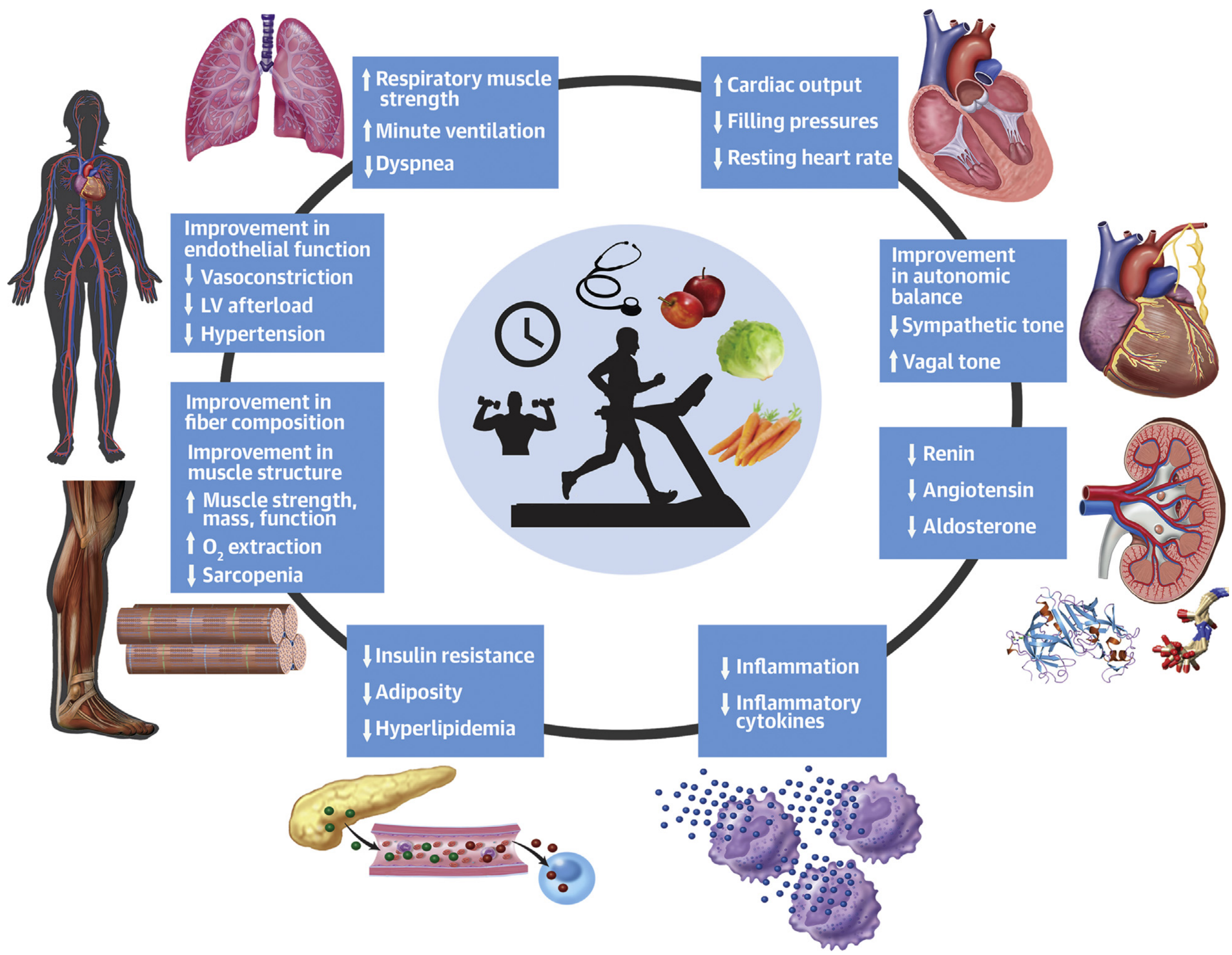
MCT = moderate continuous training

NYHA = New York Heart Association

QOL = quality of life

V_{O_2} = oxygen uptake

CENTRAL ILLUSTRATION Mechanisms of Beneficial Effects of Exercise Training and Cardiac Rehabilitation in Patients With Heart Failure



Bozkurt, B. et al. J Am Coll Cardiol. 2021;77(11):1454–69.

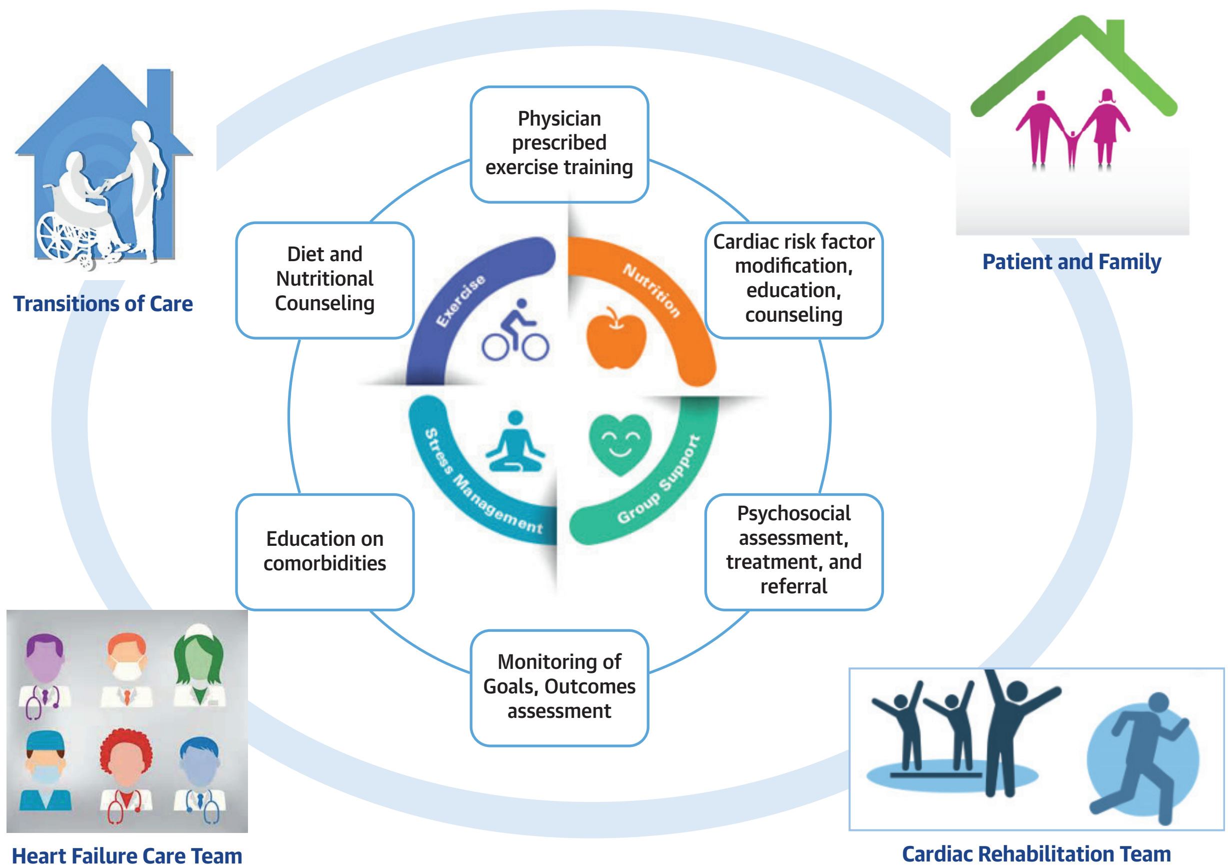
Mechanisms by which cardiac rehabilitation and exercise training improve overall status in patients with heart failure.

(**Figure 1**). Many programs include behavioral health personnel and pharmacists as integral members of the team, while others have such expertise available for referral and consultation when required.

It is important to note that in patients with cardiovascular disease, some of the beneficial effects of CR in reducing cardiovascular mortality and hospitalizations have been attributed to reductions in smoking, cholesterol, and blood pressure in addition to exercise (56,57), underlining that the key aims of CR are not only to improve physical health and QOL but also to assist people with HF to develop the necessary skills to successfully self-manage (58). Therefore, a CR program should contain specific components that optimize cardiovascular risk reduction, foster healthy behaviors and compliance, reduce disability, and promote an active lifestyle for patients with HF and cardiovascular disease (54).

The core components of a CR program include baseline patient assessment; nutritional counseling; lifestyle modification; risk factor management for lipids, blood pressure, weight, diabetes, and smoking; psychosocial interventions; and physical activity counseling and ET (52) (**Table 1, Figure 1**). Programs that consist of ET alone are not considered CR programs. Randomized controlled trials have shown that multidisciplinary comprehensive programs including self-care strategies beyond ET significantly improve

FIGURE 1 Common Components of Cardiac Rehabilitation and Collaborative Teams



Common components of cardiac rehabilitation are represented in the **central circle**. There should be close collaboration and communication among the cardiac rehabilitation team, heart failure care team, consultants, patient and family members, and other team members involved in transitions of care, especially post-discharge following hospitalizations and acute to nonacute or long-term care facilities, which is represented in the **outer circle**.

exercise capacity and reduce hospitalizations and mortality (59).

BARRIERS AND LIMITATIONS TO IMPLEMENTATION OF CR

Despite encouraging results, outcome benefits, and cost-effectiveness, CR remains underused, with participation rates ranging from 10% to 30% worldwide (18-20). Even in the highly supervised HF-ACTION clinical trial, although exercise equipment was provided at home and intense efforts were made to increase adherence, long-term adherence was <30% (1,60). These underscore the necessity of enhanced patient surveillance, closer monitoring, behavioral interventions, individualization, and adjustment according to patient symptoms and tolerance to increase adherence.

Multiple studies have looked at factors contributing to inadequate use of CR and prognostic variables to improve adherence (18,19,61-63). Three factors play key roles: 1) health care providers and the system; 2) patients; and 3) health care policy (20). A key variable accounting for barriers to adoption among providers remains lack of awareness and education about benefits of CR. It is known that patient adherence would be higher with stronger physician endorsement. Lack of trained staff members and facilities coupled with poor reimbursement and high costs also contribute to limited delivery of care. Robust education of providers and extension of provider teams beyond cardiologists to include primary care clinicians and advanced practice providers may help overcome this challenge and make it more cost effective.

TABLE 1 Key Features of a Cardiac Rehabilitation Program for Patients With Heart Failure

Baseline patient functional capacity, physical activity, and tolerance assessment
Individualized risk assessment for heart failure and comorbidities
Individualized exercise prescription
Monitored exercise (including telemetry)
Educational program
Diet and nutritional counseling
Access to smoking cessation program
Psychological evaluation and treatment as appropriate
Monitoring of individual patient and overall program goals
Comprehensive review of medications, including dosing and adherence
Communication and interaction with appropriate physicians

Multiple psychosocial, economic, and physical factors influence adherence (18,19,61-63). Studies have shown that women (64), minorities, and elderly patients are more likely to be deprived of the benefits of CR and may exhibit poor adherence (19,63). Competition of time for care of family, poor access to rehabilitation prescriptions, inadequate insurance coverage, time lost from work, poor social support, lack of perceived benefit, and travel challenges interfere with referral and adherence. Among elderly patients, overall poor functional capacity, sarcopenia, and muscle and bone loss also limit participation (18,19,61-63).

Finally, health care policies have a major impact on participation rates (20). Policy and coverage decisions may hinder novel models of rehabilitation, including those with alternative staff members, home-based and telemedicine models, and alternative durations of therapy. Costs to participate in rehabilitation and copayments in insurance plans can be prohibitive to adherence to prescription. Although the barriers are multifold and appear daunting, a combination of re-engineering of our health care system coupled with progressive policy can help overcome some of these challenges.

EVIDENCE OF BENEFIT WITH DIFFERENT EXERCISE TYPES IN PATIENTS WITH HF

AEROBIC OR ENDURANCE TRAINING. Aerobic exercise or endurance training remains the mainstay of ET and includes treadmill walking, cycling, upper body ergometry, dancing, swimming, and playing sports (54). Aerobic training has been shown to reverse left ventricular remodeling in patients with HF who are clinically stable, results in improvements in aerobic capacity and peak oxygen uptake (V_{O_2}), and modifies cardiovascular disease risk factors (65). Moderate continuous training (MCT) is the most evaluated ET modality, as it is efficient, safe, and well tolerated by patients with HF (1).

RESISTANCE TRAINING. In patients with HF, combined endurance-resistance exercise programs significantly improve submaximal exercise capacity, muscle strength, and QOL (2,3). Both aerobic and combined aerobic and resistance training are effective interventions to improve peak V_{O_2} in patients with HF, but resistance training enhances muscular strength and muscle mass to a greater extent than aerobic exercise (66). Combined training may be more effective in improving muscle strength and endurance (2,3). Strength training plays an important role in improving physical function and decreasing functional disability in patients with HF (67). Contrary to concerns, the addition of dynamic resistive exercises has been reported to result in acute and chronic adaptations such as increases in cardiac output, walking distance, exercise capacity, ventilator efficiency, and QOL, without adverse events or detrimental effects on left ventricular remodeling and N-terminal pro-B-type natriuretic peptide levels (2,66-70). Although combined training was not associated with a demonstrable benefit in left ventricular remodeling in a large meta-analysis, smaller studies have shown a beneficial effect on skeletal muscle function and/or peripheral vascular responsiveness with resistive training (65,69,70).

Certain groups of patients with HF, including older adults and women, are particularly likely to have sarcopenia and skeletal muscle changes (68,71). Resistance training may be a suitable strategy in such patients, as it increases muscle and bone mass. For example, in older adults, elastic bands can be deployed at home to target specific muscle groups, which may enhance the ability to perform activities of daily living and increase independence. However, it is important to understand that resistance training is complementary to aerobic exercise and not a substitute for it. Resistance training appears to be safe, but there are limited data from clinical trials because of patient selection, small subject numbers, and variation in exercise prescriptions. It may require a potential increase in staff-to-patient ratio because of safety and tolerability concerns.

INTERVAL TRAINING AND HIGH-INTENSITY INTERVAL TRAINING. Interval training has been proposed to be more effective than continuous exercise for improving exercise capacity in the general population. In this protocol, the patient alternates short bouts of moderate- to high-intensity exercise with longer recovery phases, performed at low or no work load. Small studies have suggested that high-intensity interval training (HIIT) is safe and superior to MCT in reversing cardiac remodeling and increasing peak V_{O_2} and aerobic capacity (72). There is emerging evidence that high-intensity intermittent exercise may also be beneficial, with improvements in physiology, QOL, and functional capacity in patients with HF (72-76).

A meta-analysis comparing HIIT with MCT found greater improvements in exercise tolerance but no significant effect on left ventricular ejection fraction (LVEF) at rest with HIIT in patients with HFrEF who were clinically stable (77). Another systematic review suggested that HIIT may result in a greater beneficial effect on peak V_{O_2} in patients with HF (78). A recent large multicenter trial comparing HIIT and MCT showed that HIIT was not superior to MCT in left ventricular remodeling or aerobic capacity (79). In these select patients, HIIT was safe, with no significant differences for withdrawal, death, and cardiac events compared with control arms (78,79). The effects of high-intensity rehabilitation on patient retention and adherence and the selection of appropriate candidates for safety will need to be addressed. A sampling of previous studies did not report patient dropout due to concerns of discomfort (80). Future implementation may require protocol standardization and tailoring to the needs of each patient.

MCT is more familiar to patients, and people in general, as it is more similar to lifestyle exercise (e.g., walking) and is suitable for patients with very low baseline fitness (e.g., <3 METs) (**Table 2**). MCT can also be easier for patients to perform in social situations (e.g., walking, biking with a family member). For patients with very low fitness, frailty, and/or sarcopenia, MCT in fact can be perceived as HIIT (**Table 2**). MCT is easier for staff members to implement, as less calculation and individual patient monitoring may be required. Given the physiological, autonomic function, and circulatory hysteresis, MCT may also likely be more suitable for those with LV assist devices or those who have recently undergone transplantation. In comparison, HIIT may result in higher levels of fitness and the same or better improvement in a variety of metabolic or muscle physiological measurements, and it can be more time efficient (77-79). HIIT, however, may require judicious patient selection and additional staff effort. The aggregate safety data appear to be more numerous for MCT than HIIT (1,54), and further research is needed to better characterize the efficacy and safety of HIIT in comparison with MCT in patients with HF.

INSPIRATORY MUSCLE TRAINING. Inspiratory muscle weakness is widespread among patients with HF (39). The etiology is multifactorial and includes mechanical and metabolic factors or oxidative stress.

TABLE 2 HIIT Versus MCT in Patients With Heart Failure

	MCT	HITT
Evidence base	++++	+++
Similarity to lifestyle exercise	+++	+
Time required	++	++++
Suitability for frailty/very low fitness	++++	++*
Staff effort	++	++++
Cardiometabolic benefit	+++	++++
Fitness achieved	+++	++++
Suitability for broad range of patients	++++	++
Safety	++++	++

*MCT may be in fact HIIT, as even low workloads may be substantial for these patients.
HIIT = high-intensity interval training; MCT = moderate continuous training.

Inspiratory muscle training is beneficial in improving respiratory muscle strength and dyspnea in patients with stable HF and respiratory muscle weakness (81). Multiple studies have demonstrated safety and additional favorable effects in both patients with HFrEF and those with HFpEF (82-84). In severely deconditioned patients, inspiratory muscle exercise may help transition to conventional rehabilitation while improving cardiorespiratory fitness and QOL (85). The addition of inspiratory muscle training to aerobic training has also been shown to reduce dyspnea, increase peak V_{O_2} and exercise time, and improve QOL (85-87). A meta-analysis comparing inspiratory muscle training with sham or control subjects showed improvements in 6-min walk distance, peak V_{O_2}, and minute ventilation in patients with HF (88). Inspiratory muscle training is starting to be included in individual rehabilitation programs with patients with HF.

LOCALIZED MUSCLE TRAINING. Although traditional CR involves whole-body exercise, significant benefit in exercise capacity has been shown even with small muscle exercises (89). Significant improvements in muscle structure, diffusive oxygen transport, and oxygen use in the muscles can be seen without increasing cardiac output. Localized muscle training may be an important type of training to improve exercise capacity in patients with HF and could be particularly useful in severely disabled patients with minimal reserve capacity (90).

HOW TO DEVELOP AN ET PRESCRIPTION AND RECOMMENDATIONS

An individualized exercise prescription should be developed on the basis of a baseline evaluation, with incorporation of the goals of the patient and the treatment team. Usually, providers in the CR program

TABLE 3 Exercise Training Program Used in the HF-ACTION Trial

Training Phase	Location	Week	Weekly Sessions	Aerobic Duration (min)	Intensity (% of HRR)	Mode of Exercise
Initial, supervised	Clinic	1–2	3	15–30	60	Walk or cycle
Supervised	Clinic	3–6	3	30–35	70	Walk or cycle
	Clinic/home	7–12	3/2	30–35	70	Walk or cycle
Maintenance	Home	13 to end of treatment	5	40	60–70	Walk or cycle

Adapted with permission from Whellan et al. (92).
HF-ACTION = Heart Failure: A Controlled Trial Investigating Outcomes of Exercise Training; HRR = heart rate reserve.

develop the patient's ET regimen, but the referring clinician should also be aware of the types, frequency, and duration of exercise for the patient (i.e., the exercise prescription). Of the randomized controlled trials of exercise-based CR, only 8% to 15% sufficiently described the exercise protocol (91), which highlights the knowledge gap for referring physicians. An exercise prescription can be developed by the CR specialist in collaboration with the HF provider and should specify the frequency, intensity, duration, and modalities of exercise. In the following paragraphs, we summarize a practical approach to the development of an exercise prescription as part of CR, which we believe would be useful for general clinicians or specialists who refer patients with HF for CR.

Baseline cardiopulmonary exercise testing can be used to determine whether patients can exercise safely, including checking for abnormal blood pressure responses, early ischemic changes, and significant arrhythmias. The usual method used for exercise testing is an extended and modified Naughton protocol using a motor-driven treadmill (92). For patients who are unable to perform an exercise test on a treadmill, a bicycle ergometer or walking can be used. During cardiopulmonary exercise testing, patients are strongly encouraged to achieve a Borg rating of perceived exertion >16 (on a scale of 6 to 20, with 16 being equivalent to effort of 85%) and a respiratory exchange ratio (the ventilatory response to exercise as a measure of effort expended: carbon dioxide production/VO_2 >1.05 or 1.10). Sites that do not have access to cardiopulmonary exercise testing can use graded exercise testing without gas exchange or other modalities such as the 6-min walk test to assess exercise capacity.

In the HF-ACTION trial, the heart rate reserve method was also used to guide exercise intensity. Peak heart rate was derived from a patient's most recent exercise test, and resting heart rate was taken after 5 min of quiet seated rest (92). The HF-ACTION ET protocol was designed such that patients begin exercising at low intensity and then increased to moderate intensity when they were able to do so. For the first 6 supervised training sessions, the training heart rate range was computed at 60% of the heart rate reserve (resting heart rate + 0.6 [peak heart rate − resting heart rate]) (92). For patients with significant or advanced HF symptoms who were unable to perform continuous exercise, training was initiated with rest periods, with a goal of 50% of heart rate reserve and 15 to 30 min of total exercise time. Training intensity was increased to a range of 60% to 70% of heart rate reserve for the rest of the supervised ET sessions and for the home-based ET phase (**Table 3**). The supervised training phase of the trial consisted of 36 supervised training sessions, with a goal of 3 sessions per week, subsequently continued with home exercises, 5 times per week (92). For the home maintenance phase, patients were asked to perform 40 min of aerobic exercise. Before and immediately after each aerobic period, patients completed a 10-min warmup period and a 10-min cooldown period. This protocol allowed patients to either cycle (using a stationary bicycle) or walk (using a treadmill or walking independently) (**Table 3**) (92). For patients who were in persistent atrial fibrillation or who had frequent ventricular ectopic beats that make exercise prescription by heart rate reserve and the use of heart rate monitors invalid, the rating of perceived exertion was used to guide exercise intensity. Patients used these techniques when exercising at home to ensure that exercise intensity is within the ranges set for them after each exercise test and during follow-up visits throughout the HF-ACTION trial (92).

Some patients may be unable to exercise up to the prescribed heart rate, such as those taking β-blockers. In these patients, a rating of perceived exertion between 12 and 14 may be used to guide exercise intensity (92). For patients who exhibit exercise-induced ischemia or angina, exercise intensity can be set at a heart rate of 10 beats/min less than the heart rate at which the onset of angina or ischemia occurs (92).

TABLE 4 Frequency, Intensity, Time, and Type of Exercise Regimen Commonly Used for Patients With Heart Failure in Cardiac Rehabilitation

	Aerobic Exercise	Resistance Exercise
Frequency	5 days/week, moderate intensity* 3 days/week, high intensity†	2 or 3 nonconsecutive days/week
Intensity	Exercise in target heart rate Focus on a variety of intensities	Determined by the amount of weight lifted and the repetitions and sets Goal of 8–10 exercises, about 1–3 sets of 8–16 repetitions of each exercise
Time	30–60 min/session; shorter if exercise is high intensity	Depends on strength and schedule: up to 1 h for total body workout, less for split-routine workout
Type	Any activity that increases heart rate, such as running, walking, cycling, or dancing	Activities using resistance: bands, dumbbells, machines, body weight exercise

Modified with permission from Josephson and Mehanna (93). *Moderate intensity: 50% to 69% of target heart rate. †High intensity: 70% to <90% of target heart rate.

A complementary model in which "traditional" aerobic ET is combined with subsequent intermittent resistance training may be suitable for select patients who can tolerate resistance training. A specific example of a protocol that incorporates resistance training may start with shorter intervals at low intensity, such as a 10-s bout of walking. As tolerance increases, both duration and intensity may be increased over 10 to 12 weeks. For severely debilitated patients, ultrashort, 5- to 20-s bursts of low- to moderate-intensity exercise interspersed with rest periods may be beneficial to improve functional capacity. As may be the case with aerobic training, resistance training can be started at low weight and higher repetitions and increased gradually. Resistive exercise can be offered using weight or therapeutic bands, which may be easier to incorporate in activities of daily living among older patients with HF. The goal of either activity should be to increase skeletal muscle aerobic activity as well as increase skeletal muscle mass. Among morbidly obese patients with HF, the use of external supports such as externally supported walking with harnesses or antigravity treadmills may be useful in overcoming some of the challenges and may aid in weight loss.

A useful framework for the exercise prescription is the acronym FITT: frequency, intensity, time, and type. For example, a patient may be prescribed an every-other-day (frequency), moderate-intensity (intensity is usually judged by rating of perceived intensity, which can be measured using the Borg scale), 20-min (time) walking (type) regimen. Similarly, a patient may be prescribed strength training as twice-weekly (frequency) 5-pound (intensity) 12-repetition (time) dumbbell biceps curls (type) (93). Simplified examples of FITT are listed in **Table 4**.

The effects of exercise prescription on functional capacity can be monitored by changes in symptoms, New York Heart Association (NYHA) functional class, QOL, functional capacity measured by changes in V_{O_2} during peak exercise by cardiopulmonary exercise testing (94), or 6-min walk distance.

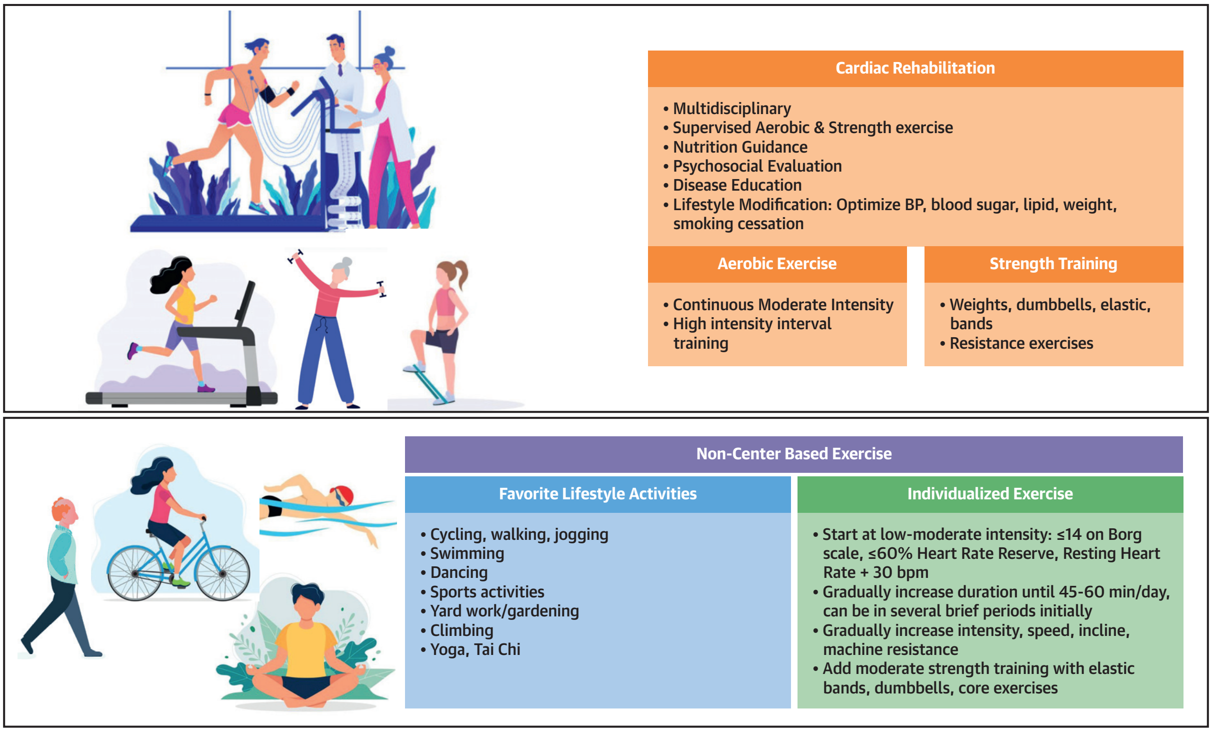
For those who are not in a formal CR program, "lifestyle exercise" (walking, taking the stairs when feasible, parking a distance from a store, light gardening, dancing) and structured activity and exercise can be helpful. The latter is typically MCT with suitable individualized activities (walking, swimming, exercise biking), the intensity of which can be guided by the Borg scale, heart rate reserve, or resting heart rate. Activity can be started at low to moderate intensity, such as <14 on the Borg scale, at <60% heart rate reserve or at resting heart rate plus 30 beats/min. The duration of exercise can be increased every 2 to 4 weeks until 45 min in total duration (e.g., 3 sessions 15 min each), and then intensity, speed, and incline can be gradually increased, guided by the Borg scale, heart rate reserve, or to a target of 85% of age-predicted maximum heart rate. Moderate strength training with elastic bands, dumbbells, and core exercises can also be added (**Figure 2**).

The Physical Activity Guidelines for Americans, adopted by many societies, recommend that adults in the general U.S. population should aim for at least 150 min per week of moderate-intensity or 75 min per week of vigorous-intensity aerobic exercise, along with at least 2 days of muscle-strengthening activity (95).

DEVELOPMENT OF THE NONEXERCISE COMPONENTS OF CR

Although exercise therapy is central to the CR process and benefit, other components are essential as well. These components may be central to the disease etiology and treatment necessary for secondary prevention (e.g., nutrition, tobacco cessation) and/or may be related to common comorbidities or sequelae (e.g., depression) (**Table 1**). Their evaluation and treatment are warranted for optimal patient benefit.

FIGURE 2 **Components of Center-Based Cardiac Rehabilitation and Non-Center-Based Exercise Activities**



Components of center-based cardiac rehabilitation program and non-center-based exercise activities appropriate for patients with heart failure. BP = blood pressure.

Diet and nutrition are key contributors to the pathophysiology of cardiovascular disease. Although it is beyond the scope of this paper to review the specifics of various diets (e.g., vegetarian, Mediterranean, Dietary Approaches to Stop Hypertension), dietary sodium intake, and weight-loss diets, it is important that this be done on an individual basis with each CR participant. It is also important to assess the adequacy of caloric intake and dietary protein, especially in patients with muscle wasting, which may be masked in patients with obesity. Emphasis on personalized dietary counseling that takes into account symptoms, congestion, and comorbidities, including the presence or absence of diabetes and atherosclerotic heart disease, is important. Typically, standardized and validated dietary questionnaires are used to assess pre-CR diet, develop individual patient goals and strategies to achieve those goals, and reassess diet at or near CR completion. Although practitioners other than dieticians can conduct these assessments, and group education sessions may be led by any qualified individual (**Table 5**), registered dietician nutritionists are the ideal personnel to perform nutritional assessment and counseling, if available. Many programs include sessions on ordering wisely from restaurant menus and may include a trip to a grocery store to provide experiential learning as well. Some patients may require more intensive or prolonged nutrition counseling and referral to a dietician.

Lifestyle modification with physical activity, avoiding obesity, not smoking, maintaining a healthy diet, reducing cholesterol, and keeping blood pressure and glucose normal is associated with a lower lifetime risk for HF and greater preservation of cardiac structure and function (96). Lifestyle modification counseling is usually included in CR.

Depression, anxiety, and mild cognitive dysfunction are common in patients with HF. It is expected that screening assessment of these conditions be part of the initial evaluation and when appropriate at intervals during the program progression. Standardized and validated instruments (e.g., the Beck Depression Inventory, the Patient Health Questionnaire) are typically used for this purpose. Those patients with severe depression and/or suicide risk should be promptly referred to qualified experts, and appropriate treatment should be initiated. For patients with mild symptoms of depression and/or anxiety, the supportive nature of most CR programs is associated with longitudinal improvement. Group education typically includes sessions on the affective components of HF and cardiovascular disease (stress, anxiety, depression, anger), which helps validate patients' concerns and provides strategies for treatment and improvement (**Table 5**). Many programs include behavioral health expert clinicians (e.g., psychologists, social workers), while others have this as a referral resource.

CR is an ideal environment also to provide much-needed and valuable patient education. It is typically done via both group education sessions and individualized patient education. Common topics include recognition of HF, self-care strategies including weight and fluid retention monitoring and diuretic agents, medication adherence, cardiac anatomy and function, cardiac procedures, cardiac devices, pharmacotherapy, and optimization of guideline-directed therapies (**Table 5**).

RECOMMENDATIONS FOR CR IN CURRENT HF GUIDELINES

CR is now recognized as integral to the care of patients with chronic, stable HF. In the current American College of Cardiology/American Heart Association guideline for the management of HF (97,98), CR is a Class 1 recommendation (Level of Evidence: A). Ambulatory symptomatic (stage C) patients with stable NYHA functional class II or III HF symptoms, on guideline-directed medical therapy, should be considered for supervised CR (53,97,98). This recommendation is supported by evidence of improvements in exercise capacity, QOL, and psychological well-being and reductions in hospital admissions with ET in patients with HF (1,10-17,99).

In the United States, the Centers for Medicare and Medicaid Services and most insurers authorize coverage for CR services for at least 6 weeks (usually 36 sessions) for patients with stable chronic HF with LVEF ≤35% and NYHA functional class II to IV symptoms despite 6 weeks of treatment with optimal HF therapy. These authorization criteria reflect the inclusion criteria of the HF-ACTION trial (1). Per coverage criteria, stability is defined as no recent (≤6 weeks) or planned (≤6 months) major cardiovascular hospitalizations or procedures. At present, there is no Centers for Medicare and Medicaid Services coverage for CR in patients with HFpEF, despite evolving evidence (10,15,16,49,100-104).

The target for patients with HF who are clinically stable is to perform aerobic medium-intensity continuous ET (i.e., 50% to 80% of peak capacity) for up to 45 min on most days of the week. The modality of exercise can be selected according to baseline functional and exercise capacity and uptitrated in duration and intensity as tolerated to this target.

EVIDENCE OF BENEFIT OF ET IN PATIENTS WITH HFrEF

ET leads to improvements in central hemodynamic status and peripheral vascular, endothelial, and skeletal muscle function (105), attenuation of sympathetic and neurohormonal activation (39,43-45), reduction in circulating levels of N-terminal pro-B-type natriuretic peptide (41), and increases in vagal tone (38,106) (**Central Illustration**).

In clinical trials, LVEF and reverse remodeling have not been measured as primary outcomes, and when reported, improvements were usually modest (50) or did not reach significance despite favorable trends (40). A meta-analysis of randomized controlled trials of ET in patients with chronic HF reported that aerobic training was associated with small but significant improvements in LVEF and left ventricular end-diastolic and end-systolic volumes (65). In another meta-analysis in patients with HFrEF who were clinically stable, ET was associated with a modest increase in LVEF, with the greatest benefits occurring with long-term (≥6 months) training. HIIT performed for 2 to 3 months also was associated with increases in LVEF, but resistance training performed alone or combined with aerobic training did not significantly change LVEF (42). Overall, ET is not associated with adverse effects on left ventricular remodeling and function, and in some patients, there may be improvements. However, given modest changes seen only in meta-analyses, LVEF or reverse remodeling should not be overemphasized as a target or marker for benefit. More important, most patients with HFrEF improve their functional capacity without improvements in LVEF, and LVEF improvement is not necessary to obtain physiological and clinical benefits from ET.

The beneficial effects of ET on skeletal muscle structure and function are also well characterized, with a significant increase in the oxidative capacity of skeletal muscle (107), correlating with changes in functional capacity and peak V_{O_2}.

Studies and meta-analyses have repeatedly shown that ET results in improvements in health-related QOL in both patients with HFrEF and those with HFpEF (11,14,17,108-111). In HF-ACTION, health status improved more during the initial 3-month period

TABLE 5 Common Topics for Group Education Sessions in Cardiac Rehabilitation

Nutrition: lipids, salt, prudent diet, weight loss, label reading, dining
Recognition and monitoring of heart failure symptoms
Self-care management strategies in heart failure, including weight monitoring and diuretics
What is heart failure? Cardiac anatomy and physiology
Cardiac terms and diagnoses
Cardiac procedures and devices commonly used in heart failure
Heart failure medications and other drugs commonly used in patients with heart failure
Self-care strategies for management of comorbidities in heart failure
Tobacco cessation
Recognition and management of psychosocial stress
Work, hobbies, physical activity beyond rehabilitation
Lifetime healthy lifestyle program
Goals of care, end of life, communication with family and providers

(60) and persisted throughout follow-up and also in key subscales, including physical limitations, social limitations, and QOL.

ET also results in improvements in exercise capacity (11,108,110), with an approximately 15% to 17% improvement in peak V_{O_2} (10). Systematic reviews have shown improvements in exercise duration, work load and 6-min walk distance along with improvements in QOL (11,14,17,108-110).

In recent meta-analyses, ET was associated with reductions in HF hospitalization and, in some reports, all-cause hospitalization (11,12,17,109). Most studies and meta-analyses have not shown significant changes in mortality (1,11,12,14,17), until recent analyses suggesting that ET may be associated with reductions in mortality, especially with longer follow-up (109,112). In the HF-ACTION trial, there reductions in mortality and hospitalizations were noted among those who achieved ET prescription targets (7).

EVIDENCE OF BENEFITS OF ET IN PATIENTS WITH HFpEF

There are emerging data on the benefits of ET in patients with HFpEF (10,15,16,49,100-104).

Several studies and meta-analyses have demonstrated significant improvements in peak and submaximal exercise capacity, cardiorespiratory fitness, and QOL in patients with HFpEF (13,49,100,101, 104,111). Although some studies showed atrial reverse remodeling and improved LV diastolic function (101), other studies and meta-analyses failed to show differences in left ventricular function, compliance, volumes, or arterial stiffness in patients with HFpEF (13,103,104,111). Overall there is concordant evidence that ET confers benefits in terms of improvements in exercise capacity and health-related QOL and appears to be safe in patients with HFpEF.

EVIDENCE OF BENEFITS OF ET FOLLOWING HOSPITALIZATION FOR ACUTE HF

The recently published randomized multicenter EJECTION-HF (Exercise Joins Education: Combined Therapy to Improve Outcomes in Newly-Discharged *HF*) trial showed that supervised center-based ET was a safe and feasible addition to disease management programs with supported home exercise in patients recently hospitalized with acute HF but did not reduce the combined endpoint of death or readmission (113). The ongoing REHAB-HF (Rehabilitation Therapy in Older Acute HF Patients) trial is a randomized trial of a physical function intervention in older patients hospitalized for HF whose aim is to demonstrate whether balance, mobility, strength, and endurance ET improves physical function and reduces rehospitalizations (114,115). Currently, limited data suggest that physical activity and ET are safe and feasible additions to disease management programs in patients who are stable after recent hospitalizations with acute HF.

EVIDENCE OF BENEFITS OF ET IN ADVANCED OR STAGE D HF

There are not enough data to assess the efficacy or safety of CR for patients with active NYHA functional class IV HF symptoms or stage D HF (116,117). In patients with advanced NYHA functional class III HF, long-term ET was associated with an improvement in stroke volume and a reduction in cardiomegaly (50). There are also studies demonstrating the benefit of ET for patients with durable mechanical circulatory support (118-120) and/or early after cardiac transplantation (121-123). The Rehab-VAD (Cardiac Rehabilitation in Patients with Continuous Flow Left Ventricular Assist Devices) trial showed that moderate intensity aerobic training was safe in patients with continuous-flow LV assist devices and yielded significantly improved health status, treadmill time, and leg strength, without significant improvements in peak V_{O_2} (124). If a patient with advanced HF can tolerate physical activity, ET can improve QOL (69,116,117,125). It is important to recognize that patients with advanced HF, including those with mechanical circulatory support, unfortunately are subject to the Centers for Medicare and Medicaid Services guidelines for CR in patients with HFrEF. It is also important to recognize that patients who require mechanical circulatory support for destination therapy are older, are more frail, and have more comorbidities than patients who require mechanical circulatory support as a bridge to transplantation, which influences the type of CR program they would need.

FUTURE DIRECTIONS: NEW MODELS OF CARE AND PROPOSED CHANGES FOR COVERAGE FOR CR

Novel strategies exist to overcome some of the barriers (Table 6) (20,126). Although the traditional CR model was based on prolonged aerobic training, it is often challenging to maintain adherence among patients with competing obligations, limited travel options, and income lost during therapy sessions. In recent years, several other care models have emerged, including home-based rehabilitation, technology-enhanced CR at home with mobile or Internet-based communication, social media platforms, self-monitoring health devices with regular nurse or other in-person visitations, community-based programs, or telemonitoring (126–133). Hybrid programs that are facility based combined with home-based models and telephone, telemedicine, and Internet-based models (134) and telemedicine CR are rapidly expanding, especially in rural areas with no physical access or during the recent coronavirus disease-2019 pandemic. Use of technology can improve patient awareness, engagement, and retention and patient and provider experience and can allow close monitoring of patients who may be geographically disadvantaged or with barriers (135). These efforts, coupled with pedometers, cellular communications, wearable technologies, global positioning devices, and activity monitoring through implanted devices, can help monitor physical activity and progress, reduce recidivism to inactivity, and help maintain benefits achieved. Newer exercise models can include strength training or shorter busts of HIIT (75,76) and inspiratory muscle training, and other modalities of fitness such as yoga and tai chi can be complementary to traditional ET (81).

TABLE 6 Strategies to Enhance Referral and Adherence of Cardiac Rehabilitation

Referral and adherence
Automatic referrals
Discharge order sets and papers
Shared decision making
Incorporating rehabilitation in reportable performance measures
Chronic disease management
Alternative models for service delivery
On-site versus remote, home-based monitoring
Technology-enhanced home-based cardiac rehabilitation
Use of kiosk or computer-based programs
Varying frequency and duration of program
Patient care coordinated through nurse case management
Alternatives to traditional exercise
Lifestyle exercise (e.g., walking, taking the stairs, gardening, dancing, swimming, cycling)
Physical activity counseling
Promoting caloric expenditure for weight loss and weight control
Flexibility, yoga, tai chi
Inspiratory muscle training
Education and technology
Remotely monitoring patients to expand services to underserved populations
Web-based technology to monitor progress and provide interventions such as behavioral weight loss and exercise counseling
Use of devices to monitor physical activity, such as accelerometers, pedometers, cellular communication, and global positioning devices

Adapted with permission from Clark et al. (126).

OVERCOMING BARRIERS TO REFERRAL

Inequalities exist in referral and access to CR (136). Older patients, women, and racial minorities are disproportionately not referred to CR (61–64). Alternative CR models, payer coverage strategies, and health policy changes will need to be catered to individual patients, health care system, and geographic needs and can help overcome some of the current disparities and underserved groups. We also propose a change in the Centers for Medicare and Medicaid Services coverage rule such that referral is not delayed for 6 weeks (51). We recognize the benefits of patients' receiving optimal medical therapy, but we believe that this approach is a major impediment to automatic referrals. Optimization of medical therapy can be done concurrently with CR, and patients in stable condition should be able to be referred to CR without a waiting period of 6 weeks. Adoption of automatic referral using electronic medical records can lead to a doubling in the number of participants enrolled in CR programs (137). Shared decision aids should be used to better educate patients to make informed decisions. Policies directed at including rehabilitation referral as a quality and publicly reported performance measure can enhance the number of referrals and guide health care systems to invest in rehabilitation and education. Payers should consider reducing or eliminating copays or reducing premiums for active patient participants and should include the new models of CR in provider reimbursement models. As Medicaid programs expand, CR will need to be included under essential services to reach more patients with HF and considered for patients with HFpEF and/or recent hospitalization. Policy changes are needed to address the provision of cost-effective options and to empower health care systems to invest in preventive therapies (20). Additionally, for centers that care for patients from a geographically

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diverse area that extends beyond a single rehabilitation center or that do not have on-site rehabilitation, maintaining an active list of CR programs within the referral catchment area would be helpful for timely referral.

CONCLUSIONS

CR is beneficial in patients with HF and is recommended as a Class 1A indication in HF practice guidelines. Clinicians, health care leaders, and payers should show attention to incorporating CR as part of standard care for patients with HF. The practical guidance, along with existing and emerging data, and strategies to improve referrals, adherence, and coverage for CR provided in this document should be helpful in the implementation of CR in the care of patients with HF.

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